RST-UNet: Medical Image Segmentation Transformer Effectively Combining Superpixel

Songhang Ye[1], Zhoule Feng[1], Guoheng Huang[1](✉), Jinghong Ke[1], Xuhang Chen[2](✉), Chi-Man Pun[3], Guo Zhong[4], and Xiaochen Yuan[5]

[1] Guangdong University of Technology, Guangzhou, China
kevinwong@gdut.edu.cn
[2] Huizhou University, Huizhou, China
[3] University of Macau, Zhuhai, China
[4] Guangdong University of Foreign Studies, Guangzhou, China
[5] Macao Polytechnic University, Macau, China

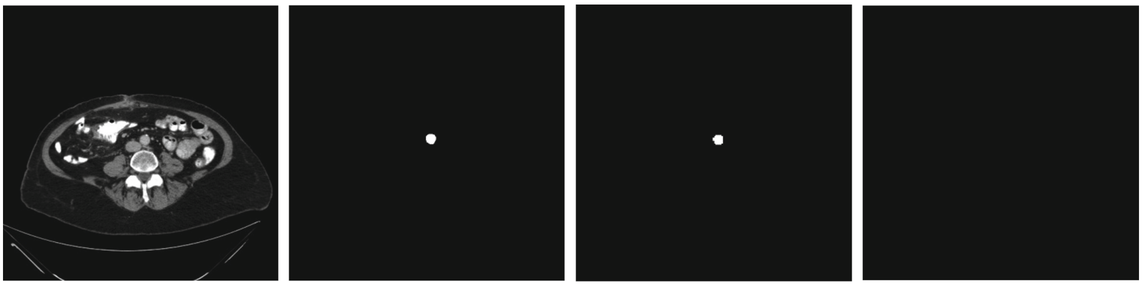
Abstract. Medical image segmentation has advanced with models like UCTransNet, TransUNet, and TransClaw U-Net, which integrate Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). However, these models face limitations due to the locality of convolutions and the computational demands of Transformers. To overcome these challenges, we introduce RST-UNet, an innovative encoder-decoder network that balances effectiveness with computational efficiency. RST-UNet features two groundbreaking innovations: the Compact Representation Block (CRB) and the Compact Dependency Modeling Block (CDMB). The CRB utilizes superpixel pooling to capture long-range dependencies while minimizing parameters and computation time. The CDMB integrates superpixel unpooling with attention mechanisms and Rotary Position Embedding (RoPE) to enhance long-range dependency modeling. This approach emphasizes critical regions and leverages RoPE to capture extensive image dependencies effectively. Our experimental results on publicly available synapse datasets highlight RST-UNet's exceptional performance, particularly in segmenting small organs such as the gallbladder, right kidney, and pancreas. Remarkably, RST-UNet achieves superior results without pre-training, showcasing its high adaptability for diverse medical image segmentation tasks. This work represents a significant advancement in developing efficient and effective algorithms for medical image analysis.

Keywords: TransUNet · Superpixel · Medical image segmentation · Rotary Position Embedding

1 Introduction

Segmentation plays a crucial role in identifying and labeling various tissues and organs, supporting tasks such as lesion detection, surgical planning, and a range

M. Mahmud et al. (Eds.): ICONIP 2024, CCIS 2289, pp. 336–350, 2025.
https://doi.org/10.1007/978-981-96-6969-1_23



(a) (b) (c) (d)

Fig. 1. (a) Sections of an abdominal CT scan. Predictions by (b) Ground Truth, (c) RST-UNet, and (d) TransUNet. In Fig (d) TransUNet, it is shown that the small organ aorta is not segmented out due to the lack of ability to model long-range dependencies, whereas our model based on rotational position encoding and superpixels can correctly capture this information.

of diagnostic and therapeutic procedures [5,8,11,16–19,26,45]. Segmenting multiple organs presents significant challenges due to their varying sizes and distinct features, often necessitating the use of manual or semi-automatic methods to achieve precise results.

The advent of Fully Convolutional Networks (FCN) [30] revolutionized semantic segmentation with their U-shaped network structure, which recovers feature map sizes to match the input image and improves segmentation accuracy through skip connections and deconvolution operations. Building on FCN, the U-Net [25,34,38,42] network enhances detail preservation with its symmetric encoding-decoding structure and skip connections, significantly improving segmentation performance and achieving considerable success.

However, these approaches face challenges such as the limitations of convolutional operations, high computational costs, inefficiency with large data volumes, and difficulties in modeling long-range dependencies and global context. To address these issues, researchers have explored feature pyramid methods [3,14,22] and CNN-based self-attention mechanisms, but these approaches still struggle with capturing long-range dependencies effectively.

Building on the success of the Transformer [40] in natural language processing, researchers introduced the Vision Transformer to tackle image and video understanding tasks [9,24,27,28]. It models long-range dependencies effectively by feeding 2D image blocks with positional embeddings into the Transformer encoder. After large-scale pre-training, these models transfer well to downstream tasks. Examples include Swin-Transformer [29], which uses window-based self-attention and a hierarchical structure to reduce computational effort while capturing multi-scale features. TransUNet [2] incorporates a Transformer structure to enhance the modeling of long-range dependencies. These models demonstrate significant capability for image segmentation.

However, most of these models use absolute or relative positional embedding, which are limitations when modeling long-range dependencies. To better understand the importance of long-range dependencies, we conducted a visualization study (Fig. 1). This study highlights the need for networks to distinguish between background, mask, and foreground pixels efficiently, especially in images with scattered foreground or large background masks. Our analysis reveals that existing models like TransUNet sometimes incorrectly segment foreground into background regions, underscoring the importance of capturing long-range dependencies.

Inspired by the application of superpixel and attention mechanisms in segmentation tasks, we propose RST-UNet, a novel transformer model for medical image segmentation. To efficiently handle depth and long-range dependencies between features, we introduce the Compact Representation Block (CRB). Unlike traditional SLIC, which segments images into superpixels followed by pixel pooling, we employ a superpixel pooling strategy. This approach organizes image information into a hierarchical structure by performing pooling operations directly at the superpixel level, reducing redundant computations while preserving key contextual information. Additionally, we introduce the Compact Dependency Modeling Block (CDMB) to fully utilize dependencies from CRB. CDMB combines superpixel and Rotary Position Embedding (RoPE) to compactly represent the feature map and employs Transformer for modeling long-range dependency, thereby enhancing segmentation performance.

This paper makes the following key contributions:

1. A novel encoder-decoder network, RST-UNet, which effectively combines the advantages of CNN and Vision Transformers with powerful feature extraction and long-range dependency modeling capabilities.
2. The Compact Representation Block (CRB), which effectively extracts data information by segmenting and pooling images at the superpixel level, simultaneously reducing model parameters.
3. The Compact Dependency Modeling Block (CDMB), which combines superpixel pooling and RoPE to enhance long-distance dependency modeling while reducing the bias of absolute or relative position encoding in Vision Transformers.
4. Experimental results on synapse datasets demonstrating the effectiveness and advantages of RST-UNet over other state-of-the-art methods.

2 Related Works

2.1 Transformer in Medical Image Segmentation

Recent studies have demonstrated the significant performance enhancement of medical image segmentation models through the incorporation of Transformer technology. TransUNet [2] integrates a hybrid CNN-Transformer encoder with the traditional U-Net architecture, processing image blocks through a Transformer layer and employing multi-head attention to capture long-distance dependencies. UNETR [13] extends this approach to 3D medical image segmentation, while TransClaw U-Net [1] utilizes self-attention mechanisms in the encoder for efficient global feature learning. However, these models face computational challenges when processing low-resolution feature maps through Transformers.

Our proposed RST-UNet addresses this limitation by introducing the Compact Representation Block (CRB), which reduces the data input to the Transformer layer. Furthermore, we replace traditional absolute positional embedding with Rotational Position Embedding (RoPE), enhancing segmentation accuracy while significantly reducing computational requirements. These innovations mark a key advancement in applying Transformers to medical image segmentation.

2.2 Superpixel in Transformer

Superpixel pooling has been shown to improve semantic segmentation across various studies [36,41]. Lin et al. [23] integrated superpixel information into the Transformer layer using a superpixel pooling-inverse pooling technique, enhancing breast tumor segmentation accuracy. Zhu et al. [47] processed high-resolution images by over-segmenting them into superpixel blocks, enabling global self-attention and providing comprehensive contextual information.

Inspired by these techniques and the RoFormer (Rotational Position Embedding Transformer), our approach employs superpixel techniques to focus the model on key regions. By feeding superpixel-pooled images into a Vision Transformer (ViT) with Rotational Position Embedding (RoPE), we efficiently capture dependencies across large regions, improving the model's ability to handle long-range dependencies. This innovative approach offers a promising avenue for enhancing precision and speed of medical image segmentation.

2.3 Position Embedding

Positional embedding plays a crucial role in Transformer models, preserving spatial information of image blocks. While models like ViT and TransUNet use absolute positional embedding, Swin Transformer achieves cross-window connectivity by alternating window positions between layers. However, traditional positional embeddings face challenges with long-distance dependencies.

To address this limitation, we implement Rotary Position Embedding (RoPE), which integrates relative position information into the self-attention mechanism. This approach, also adopted by the LLaMA model [39], is particularly effective for handling long-range dependencies between pixels in image segmentation tasks. RoPE significantly enhances deep learning models' ability to understand and process complex data dependencies, especially in image processing applications, leading to improved overall model performance.

3 Method

3.1 Overview

Figure 2 provides a detailed view of the architecture for our proposed RST-UNet, which effectively integrates the advantages of superpixels and vision transformers

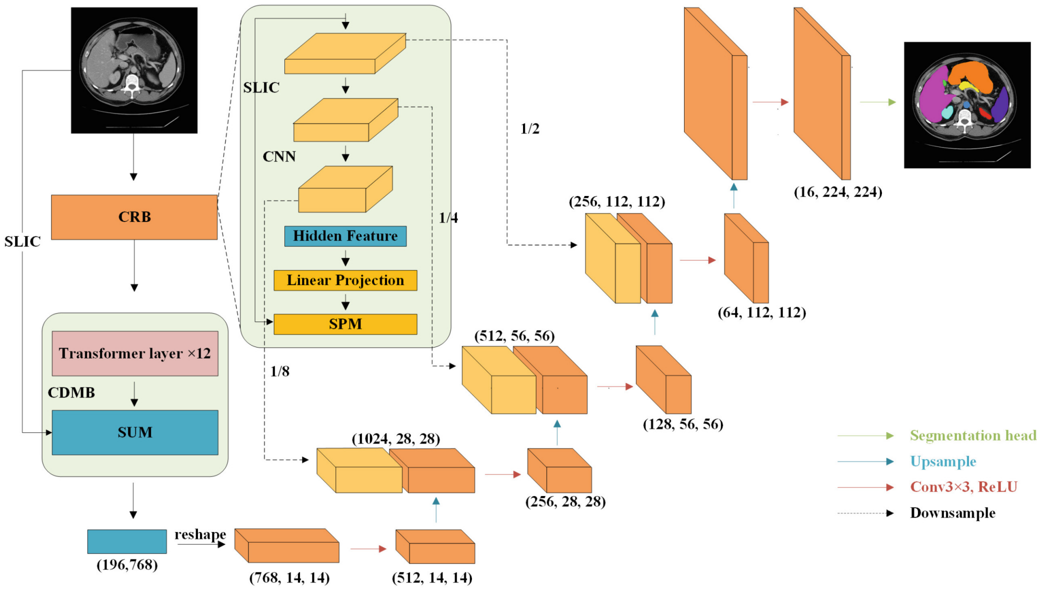


Fig. 2. Overview of the proposed RST-UNet framework.

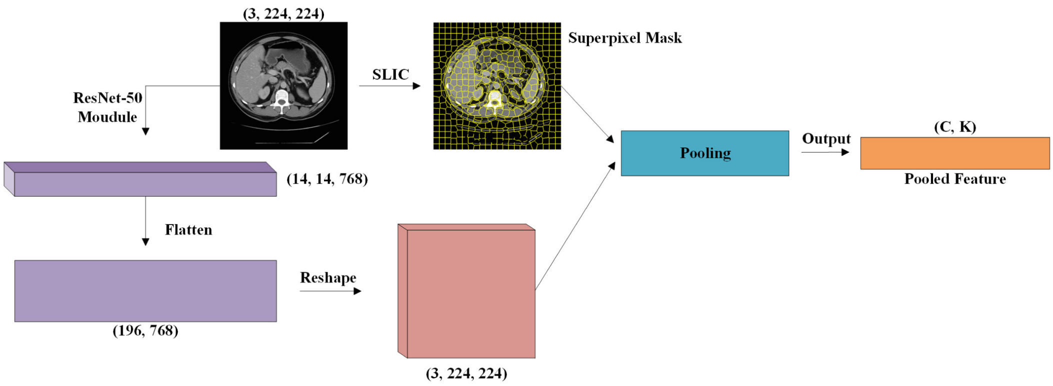


Fig. 3. General view of Compact Representation Block (CRB). Superpixel Mask and Pooled Feature are the output of the CRB.

(ViTs) to overcome limitations in capturing long-range dependencies. The network features two key components: the Compact Representation Block (CRB) and the Compact Dependency Modeling Block (CDMB). The CRB enhances model efficiency by pooling images at the superpixel level, reducing parameters and computation time. The CDMB improves segmentation performance by combining superpixel inverse pooling and attention mechanisms with rotational positional embedding (RoPE). The following subsections detail the CRB and CDMB components based on the flowchart.

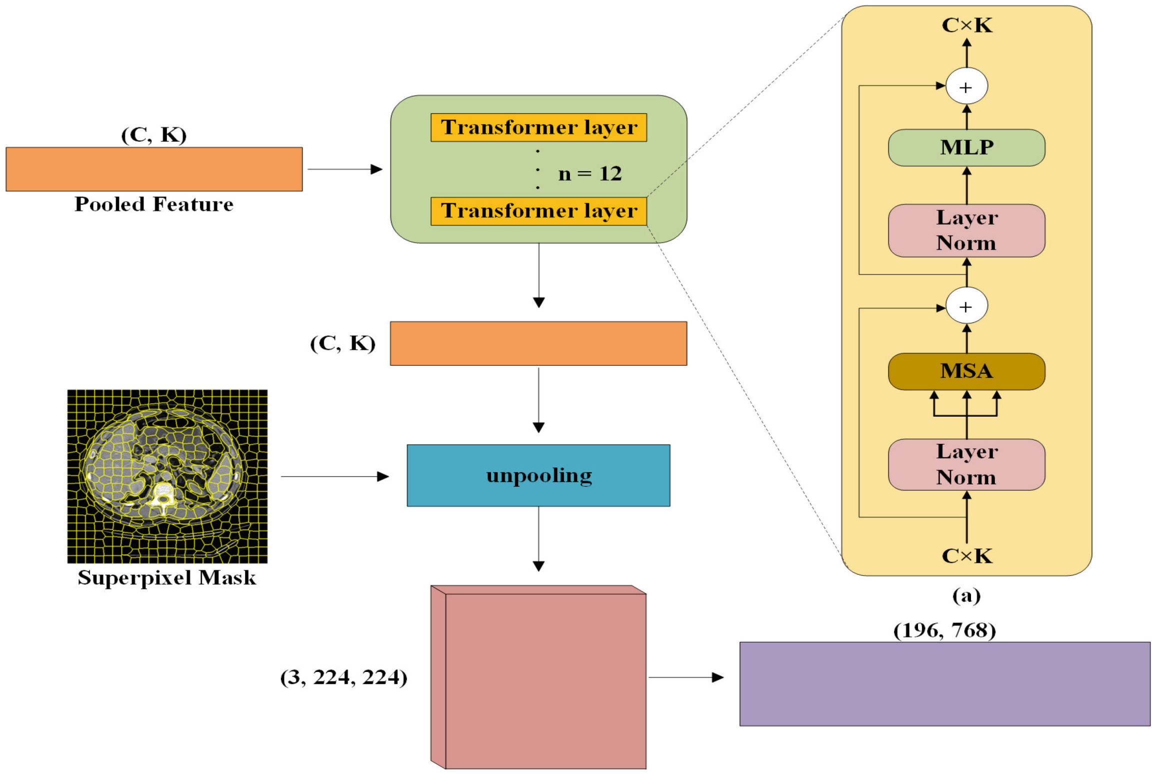


Fig. 4. General view of Compact Dependency Modeling Block (CDMB). (a) Schematic of the transformer layer.

3.2 Compact Representation Block

Modern segmentation networks focus on accurately classifying every pixel in an image. However, continuous convolution and downsampling operations may lead to loss of spatial information, decreasing segmentation performance [33]. To tackle this problem, we propose the Compact Representation Block (CRB), as shown in Fig. 3. CRB introduces superpixels to capture spatial information and refines feature maps through superpixel pooling, reducing subsequent computational requirements while preserving and concentrating key contextual information.

Superpixel Pooling Module(SPM). Superpixel pooling [36] aggregates information in the superpixel space, combining values within a superpixel block into a single value. This operation reduces image complexity and feature dimensionality while retaining crucial segmentation information. A feature map $F \in \mathbb{R}^{C \times H \times W}$ and a superpixel mask $B \in \mathbb{S}^{H \times W}$ serve as the input, where $S \in [1, K]$ denotes the integer label for each superpixel. The superpixel pooling is then performed as follows:

$$P_{C,K} = pooling\{F_{C,i} | i{:}B_i = K\} \tag{1}$$

where pooling is a function such as max pooling or average pooling, and $B_i = K$ indicates that pixel i belongs to the k_{th} superpixel. The shape of the pooled feature P is $C \times K$, where C represents the number of channels and K represents the number of superpixels.

The gradients for backpropagation are computed as follows:

For average pooling:

$$\frac{\partial P_{C,K}}{\partial F_{C',i}} = \begin{cases} \frac{1}{N(B_i)} & \text{if } B_i = K \text{ and } C = C' \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

where $N(B_i)$ denotes the total number of pixels in the k_{th} superpixel. The gradient calculation depends on the number of pixels in each superpixel. For the kth superpixel with N pixels, the gradient for each pixel i is evenly distributed across all pixels in the superpixel.

For max pooling:

$$\frac{\partial P_{C,K}}{\partial F_{C',i}} = \begin{cases} 1 & \text{if} \quad i = i^* \quad \text{and} \quad C = C' \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

In this case, only pixels chosen as the maximum value during max pooling receive the gradient. If pixel i is selected as the maximum, its gradient is 1; otherwise, it is 0.

The pooled feature is then input into the CDMB, and the resulting output, together with the mask map, is utilized in the superpixel unpooling layer.

3.3 Compact Dependency Modeling Block

To overcome limitations in capturing long-distance dependencies and reduce parameter count, we propose the Compact Dependency Modeling Block (CDMB), inspired by superpixel techniques and rotated positional coding [37]. CDMB efficiently utilize the pooled features extracted by the CRB. The CDMB process involves pooled features input to the Transformer layer, followed by inverse pooling based on the superpixel mask.

Superpixel Unpooling Module(SUM). The unpooling operation, as mentioned in [23], assigns the pooled feature to all pixels in the corresponding superpixel, restoring the feature map size. Superpixel unpooling operates as follows:

$$I_{C,i} = \{P_{C,K} | i \colon M_i = K\} \tag{4}$$

The generated feature map $P \in I^{H \times W}$ with $H = 224$ and $W = 224$ is then reshaped and input to the decoder part.

Transformer Layer. Figure 4(a) demonstrates that the Transformer layer includes multi-head self-attention mechanisms [4,6,12,20,21,31,40,44,46], a Multilayer Perceptron, and layer normalization modules. To address the challenge of capturing spatial relationships and local context in images, we incorporate rotated positional embedding, which improves the understanding of both overarching and fine-grained structures and handles long-range dependencies more effectively.

Superpixels with Rotary Position Embedding. We combine each element x_i in the superpixel feature map with its corresponding absolute positional embedding, representing the positional embedding of each element as a rotation matrix $R(p_i)$, where p_i is the positional information of the ith superpixel element. The relative position information is fused into each superpixel element by RoPE:

$$x_i^{'} = R(p_i) \cdot x_i, \tag{5}$$

where $x_i^{'}$ is a feature representation that fuses relative and absolute position information.

The attention weights $A_{i,j}$ for any two features $x_i^{'}$, $x_j^{'}$ in the sequence are calculated as:

$$A_{ij} = \frac{\exp(sim(x_i^{'}, x_j^{'})/\sqrt{D})}{\sum_{k=1}^{N} \exp(sim(x_i^{'}, x_k^{'})/\sqrt{D})}, \tag{6}$$

where $sim(x, y)$ denotes the similarity between x and y, usually computed by the dot product $x^T y$, and D is the feature vector dimension.

This approach, incorporating RoPE and superpixel techniques, enables the Transformer model to more effectively understand and capture relationships between different regions in an image, especially when dealing with long-distance dependencies and detailed features. Our method improves accuracy and performance in segmentation by reducing the data processing load while preserving sensitivity to essential image features and spatial location information.

3.4 Objective Function

To boost both pixel-wise accuracy and overlap measurement, we utilize a robust loss function that integrates Binary Cross-Entropy (BCE) with Dice Loss, as described below:

$$\mathcal{L}_{total} = 0.5 \cdot \mathcal{L}_{BCE} + 0.5 \cdot \mathcal{L}_{Dice}, \tag{7}$$

where $\mathcal{L}_{BCE}$ and $\mathcal{L}_{Dice}$ represent the BCE and Dice loss components, respectively. These are defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{j=1}^{N} \left[y_j \cdot \log(p_j) + (1 - y_j) \cdot \log(1 - p_j) \right], \tag{8}$$

$$\mathcal{L}_{Dice} = 1 - \frac{2 \cdot \sum_{j=1}^{N} y_j p_j}{\sum_{j=1}^{N} (y_j + p_j)}. \tag{9}$$

In these equations, N denotes the total number of pixels in the input sample, y_j represents the ground truth of the j-th pixel (binary value of 0 or 1), and p_j is the predicted probability for the j-th pixel.

This combined loss function takes advantage of the complementary strengths of BCE and Dice loss. BCE provides a pixel-wise measure of classification accuracy, while Dice loss addresses class imbalance by focusing on the overlap between predicted and ground truth segmentations. The equal weighting (0.5) of both components ensures a balanced optimization objective.

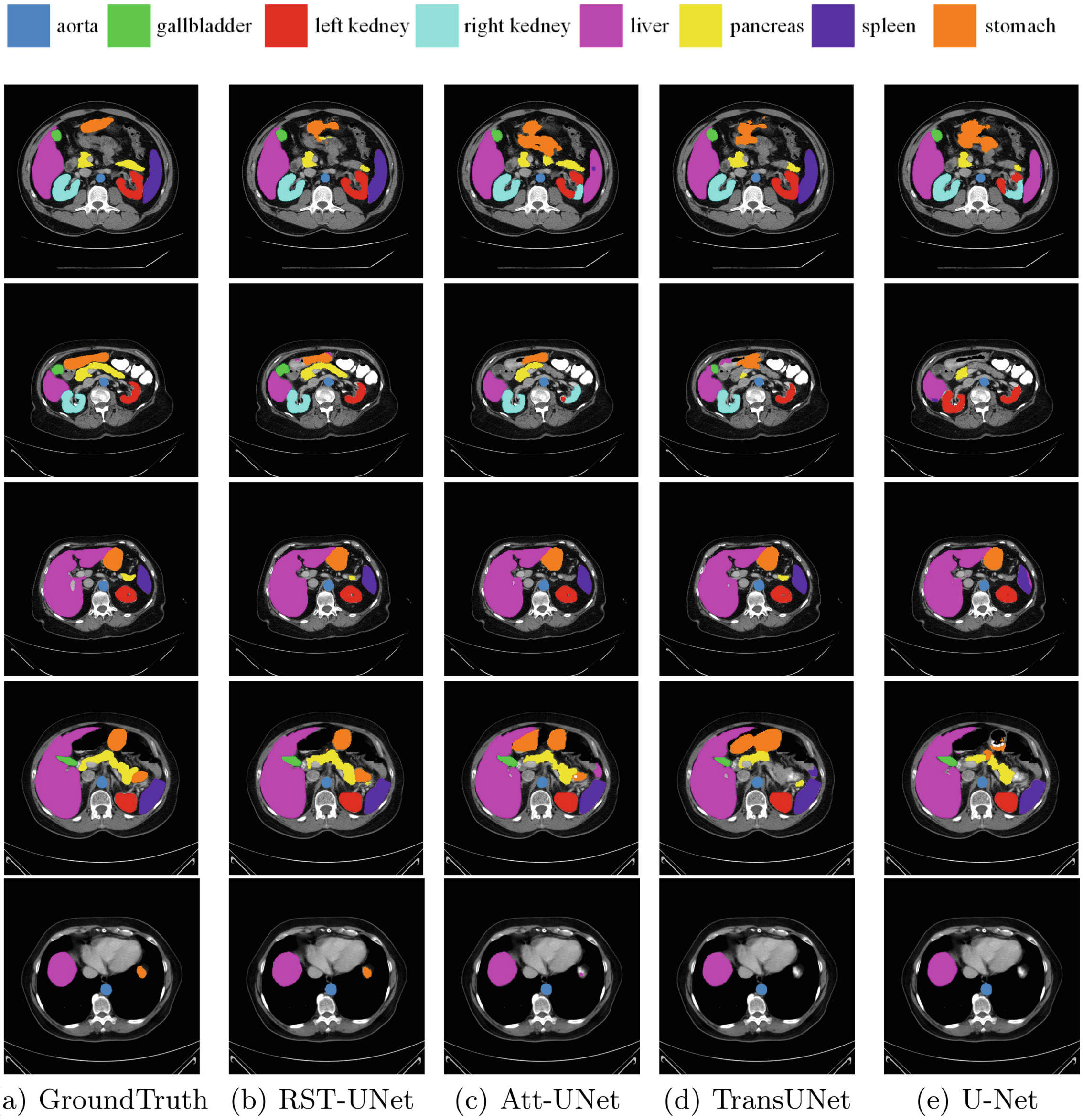


Fig. 5. Compare the results of different methods through visualization. From left to right: (a) Ground truth, (b) RST-UNet, (c) Att-UNet, (d) TransUNet, (e) U-Net, Compared with the above methods, our proposed model RST-UNet produces better results than the aforementioned models due to the characteristics introduced in this article.

4 Experiments

4.1 Datasets

We utilized the Synapse dataset, consistent with TransUNet, which comprises 30 abdominal CT scans, encompassing a total of 3,779 axially-enhanced clinical CT images of the abdomen. The voxel dimensions are 512 $\times$ 512 pixels across 85–198 slices, with field-of-view ranges from 280 $\times$ 280 $\times$ 280 mm^3 to 500 $\times$ 500 $\times$ 650 mm^3. Following the methodology of Fu et al. [10], we calculated the mean Dice similarity coefficient (DSC) and mean Hausdorff distance (HD) for eight abdominal organs: aorta, gallbladder, spleen, left kidney, right kidney, liver, pancreas, and stomach. The dataset was split into 18 samples for training and 12 for testing.

The ACDC dataset consists of 100 MRI scans from different patients, each labeled with three cardiac structures: left ventricle (LV), right ventricle (RV), and myocardium (MYO). Adhering to Chen et al. [2], we allocated 70 examples (1,902 axial slices) for training, 10 for validation, and 20 for testing.

4.2 Implementation Details

For all experiments, we employed simple data augmentation techniques, including random rotations and flips. In experiments involving the Vision Transformer [9], we used 12 layers with 16 heads. Notably, unlike many CNN and Transformer-based methods, we did not utilize any pre-training. The input resolution was configured to 224×224, with a patch size of 16. Following ResNet-50 [15] and patch embedding, we reshaped the images and performed superpixel-pooling. We used ADAM as the optimizer.

Table 1. Comparison on the Synapse dataset.

Methods	Average		Aotra	Gallbladder	Kidney(L)	Kidney(R)	Liver	Pancreas	Spleen	Stomach
	DSC ↓	HD95 ↓								
V-Net [32]	68.81	-	75.34	51.87	77.10	**80.75**	87.84	40.05	80.56	56.98
DARR [10]	69.77	-	74.74	53.77	72.31	73.24	94.08	54.18	**89.90**	45.96
U-Net [34]	76.85	39.70	89.07	**69.72**	77.77	68.60	93.43	53.98	86.67	75.58
R50 U-Net	74.68	36.87	84.18	62.84	79.19	71.29	93.35	48.23	84.41	73.92
Att-UNet [35]	77.77	36.02	**89.55**	68.88	77.98	71.11	93.57	58.04	87.30	**75.75**
R50 Att-UNet	75.57	36.97	55.92	63.91	79.20	72.71	93.56	49.37	87.19	74.95
R50 ViT [9]	71.29	32.87	73.73	55.13	75.80	72.20	91.51	45.99	81.99	73.95
TransUNet [2]	77.48	31.69	87.23	63.13	**81.87**	77.02	**94.08**	55.86	85.08	75.62
RST-UNet	**79.35**	**30.42**	87.71	68.05	<u>81.02</u>	<u>78.29</u>	<u>94.06</u>	**63.24**	<u>88.10</u>	74.34

4.3 Comparisons with State-of-the-Art Methods

To evaluate our method's effectiveness, we conducted principal experiments on the Synapse multi-organ segmentation dataset. We compared the results of our RST-UNet with previous state-of-the-art methods, including V-Net [32], U-Net [34], DARR [10], R50 U-Net, Att-UNet, R50 Att-UNet [35], R50 ViT [9], and TransUNet [2]. R50 U-Net, R50 AttnUNet, R50 ViT, and TransUNet employ ResNet50 as the backbone and were initialized with weights pre-trained on ImageNet [7]. To ensure fairness, all experimental training settings were kept consistent in this section.

Table 1 presents the experimental results on the Synapse dataset. Compared to TransUNet, our RST-UNet model achieved a 1.87% improvement in average DSC and a reduction of 1.27 mm in Hausdorff distance. This improvement underscores the effectiveness of our combined superpixel and rotational embedding approach. Unlike TransUNet, which requires ResNet50 as the backbone, our RST-UNet can be trained without pre-training, indicating better generalizability and excellent performance on medical segmentation tasks and data.

Figure 5 provides a visualization of the segmentation results for the Synapse dataset. It is evident that Att-UNet and U-Net are more prone to segmenting organs into the background or incorrectly. For instance, in the first row, both Att-UNet and UNet incorrectly segment the left kidney as the right kidney. In the second row, they incorrectly segment the gallbladder into the background, while RST-UNet avoids these errors. In the fifth row, all models except RST-UNet incorrectly segment the stomach into the background. This evidence suggests that our method has stronger long-range dependency modeling capabilities and better captures inter-pixel relationships. Comparing transformer-based models, TransUNet predicts more false-negative results compared to our RST-UNet, demonstrating RST-UNet's superior ability to model global context.

The experimental results on the ACDC dataset are presented in Table 3, showcasing the model's segmentation accuracy. These results demonstrate its powerful long-range modeling capabilities and indicate excellent generalization ability and robustness.

Table 2. Ablation study on the number of K in SLIC and we adopt TransUNet as the baseline. where when K = 700, 800, their actual number of superpixels is the same due to the limitation of the SLIC algorithm, and similarly for K = 900, 1000.

K	Average DSC	Aotra	Gallbladder	Kidney(L)	Kidney(R)	Liver	Pancreas	Spleen	Stomach
-	77.48	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
100	75.39	86.94	55.36	77.66	76.40	93.55	58.11	87.29	67.82
200	76.64	86.79	58.99	80.41	74.53	93.38	62.26	85.63	71.10
300	77.73	87.37	62.79	82.65	78.87	92.70	62.64	86.06	68.74
400	78.48	88.12	63.96	81.53	78.25	93.80	62.66	86.74	72.74
500(ours)	**79.35**	87.71	68.05	81.02	78.29	94.06	63.24	88.10	74.34
600	77.22	88.36	62.28	80.05	75.65	94.13	57.12	86.97	73.19
700/800	77.83	88.75	59.36	83.29	77.70	93.86	60.02	88.07	71.62
900/1000	78.49	87.72	67.52	83.30	78.63	93.75	59.67	87.36	69.99

4.4 Ablation Studies

To comprehensively evaluate the RST-UNet model, we performed ablation experiments on the Synapse dataset, using TransUNet as the baseline method. We verified the performance of our model under different settings through multiple ablation experiments.

The Number of K in SLIC. The input images are segmented into K superpixels. According to Zhao et al. [43], a higher number of superpixels results in smaller regions, each containing less spatial structural information of the organ. Conversely, as the superpixel region area increases, boundary structural information of organs with unclear boundaries or smaller sizes may be lost. To find the optimal number of superpixels, we adjusted K from 100 to 1000 in steps of 100 and recorded the average DSC performance for eight testing organs, as detailed in Table 2. The results indicate that our model achieves the best average DSC and HD when K = 500, which we adopted as the model setting.

Table 3. A comparison with leading methods on the ACDC dataset is presented, while other experimental results are sourced from TransUNet.

Framework	Average DSC	RV	Myo	LV
R50-U-Net	87.55	87.10	80.63	94.92
R50-AttnUNet	86.75	87.58	79.20	93.47
R50-ViT-CUP	87.57	86.07	81.88	94.75
TransUNet	89.71	88.86	84.53	95.73
RST-UNet	90.25	89.13	85.77	95.85

The Effectiveness of CDMB. We proposed the Compact Dependency Modeling Block (CDMB) for medical image segmentation to reduce the number of model parameters, better model long-range dependencies, and improve model generalization. We introduced Rotary Position Embedding (RoPE) for the first time in medical image segmentation, replacing the absolute position embedding in the transformer layer. RoPE encodes position information into embedding vectors through rotations, allowing for better differentiation of relative positions between pixels in a self-attentive mechanism to model long-range dependencies more effectively.

To verify the impact of the proposed architecture, we conducted further experiments, as detailed in Table 4. We trained two models for comparison with the baseline: one with RoPE replacing the position embedding in the transform layer, and another with no changes to the transform layer. Both used 500 superpixels and no pre-training. The results in Table 4 demonstrate that the combination of rotated positional embedding and superpixels outperforms both absolute positional embedding and superpixel combination alone. Notably, segmentation results for small organs such as the gallbladder, right kidney, and pancreas are superior with rotational position embedding compared to absolute position embedding. This indicates that our model excels in segmenting small organs, further validating the effectiveness of our approach.

Table 4. In the ablation study focused on positional embedding, we conducted experiments with the superpixel count fixed at K = 500.

Positional Embedding	Average DSC	Aotra	Gallbladder	Kidney(L)	Kidney(R)	Liver	Pancreas	Spleen	Stomach
Absolute Positional Embedding	78.01	87.74	63.10	82.12	77.94	93.70	60.01	87.08	72.37
Rotary Position Embedding	79.35	87.71	68.05	81.02	78.29	94.06	63.24	88.10	74.34

5 Conclusion

We present RST-UNet, a novel model aimed at reducing computational complexity and enhancing long-range dependency modeling in medical image segmentation. Our approach uses Rotary Position Embedding (RoPE) to capture dependencies across large image regions while maintaining spatial structure and relative position information between superpixels. We introduce the Compact Dependency Modeling Block (CDMB), which reduces computational complexity through superpixel pooling, allowing the model to focus on critical regions and better handle long-range dependencies. This makes RST-UNet particularly effective in segmenting smaller, diffuse organs compared to other advanced models. Currently, we use a fixed number of superpixels per image, which might limit performance. Future work could involve adaptive superpixel segmentation to further enhance the model's capabilities. Despite improvements, there may still be limitations at organ edges, and future work will explore fine-grained boundary detection to improve accuracy. In conclusion, RST-UNet significantly advances medical image segmentation, offering improved efficiency and accuracy, especially for challenging organ segmentation tasks. Our work lays the groundwork for future research in adaptive segmentation and robust long-range dependency modeling in medical imaging.

References

1. Chang, Y., Menghan, H., Guangtao, Z., Xiao-Ping, Z.: Transclaw U-Net: claw U-Net with transformers for medical image segmentation. arXiv (2021)
2. Chen, J., et al.: TransUNet: transformers make strong encoders for medical image segmentation. arXiv (2021)
3. Chen, L.C., Papandreou, G., Kokkinos, I., Murphy, K., Yuille, A.L.: DeepLab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. IEEE TPAMI **40**(4), 834–848 (2017)
4. Chen, X., Cun, X., Pun, C.M., Wang, S.: ShaDocNet: learning spatial-aware tokens in transformer for document shadow removal. In: ICASSP, pp. 1–5 (2023)
5. Chen, X., Lei, B., Pun, C.M., Wang, S.: Brain diffuser: an end-to-end brain image to brain network pipeline. In: PRCV, pp. 16–26 (2023)
6. Chen, X., Pun, C.M., Wang, S.: MedPrompt: cross-modal prompting for multi-task medical image translation. arXiv (2023)
7. Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L.: ImageNet: a large-scale hierarchical image database. In: CVPR, pp. 248–255 (2009)

8. Dong, Y., Chen, X., Shen, Y., Ng, M.K.P., Qian, T., Wang, S.: Multi-modal mood reader: pre-trained model empowers cross-subject emotion recognition. arXiv (2024)
9. Dosovitskiy, A., et al..: An image is worth 16×16 words: transformers for image recognition at scale. In: ICLR (2021)
10. Fu, S., et al.: Domain adaptive relational reasoning for 3D multi-organ segmentation. In: Martel, A.L., et al. (eds.) Medical Image Computing and Computer Assisted Intervention – MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part I, pp. 656–666. Springer International Publishing, Cham (2020). https://doi.org/10.1007/978-3-030-59710-8_64
11. Gong, C., et al.: Generative AI for brain image computing and brain network computing: a review. Front. Neurosci. **17**, 1203104 (2023)
12. Guo, X., Chen, X., Luo, S., Wang, S., Pun, C.M.: Dual-hybrid attention network for specular highlight removal. In: ACM MM (2024)
13. Hatamizadeh, A., et al.: UNETR: transformers for 3D medical image segmentation. In: WACV, pp. 574–584 (2022)
14. He, K., Zhang, X., Ren, S., Sun, J.: Spatial pyramid pooling in deep convolutional networks for visual recognition. IEEE TPAMI **37**(9), 1904–1916 (2015)
15. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: CVPR, pp. 770–778 (2016)
16. Huang, G., Chen, X., Shen, Y., Wang, S.: Mr image super-resolution using wavelet diffusion for predicting Alzheimer's disease. In: BI, pp. 146–157 (2023)
17. Jiang, H., Li, S., Liu, W., Zheng, H., Liu, J., Zhang, Y.: Geometry-aware cell detection with deep learning. Msystems **5**(1), 10–1128 (2020)
18. Jiang, H., Tang, S., Liu, W., Zhang, Y.: Deep learning for COVID-19 chest CT (computed tomography) image analysis: a lesson from lung cancer. Comput. Struct. Biotechnol. J. **19**, 1391–1399 (2021)
19. Jiang, H., Chen, X., Jin, C., Wang, S.: Structural brain network generation via brain denoising diffusion probabilistic model. In: AIiH, pp. 264–277 (2024)
20. Li, Z., Chen, X., Guo, S., Wang, S., Pun, C.M.: WavEnhancer: unifying wavelet and transformer for image enhancement. J. Comput. Sci. Technol. **39**(2), 336–345 (2024)
21. Li, Z., Chen, X., Pun, C.M., Cun, X.: High-resolution document shadow removal via a large-scale real-world dataset and a frequency-aware shadow erasing net. In: ICCV, pp. 12449–12458 (2023)
22. Li, Z., Chen, X., Wang, S., Pun, C.M.: A large-scale film style dataset for learning multi-frequency driven film enhancement. In: IJCAI, pp. 1160–1168 (2023)
23. Lin, T., et al.: Improving breast tumor segmentation via shape-wise prior-guided information on cone-beam breast CT images. Phys. Med. Biol. (2023)
24. Liu, W., Cun, X., Pun, C.M.: DH-GAN: image manipulation localization via a dual homology-aware generative adversarial network. Pattern Recogn. (2024)
25. Liu, W., Cun, X., Pun, C.M., Xia, M., Zhang, Y., Wang, J.: CoordFill: efficient high-resolution image inpainting via parameterized coordinate querying. In: AAAI, vol. 37, pp. 1746–1754 (2023)
26. Liu, W., Juhas, M., Zhang, Y.: Fine-grained breast cancer classification with bilinear convolutional neural networks (BCNNs). Front. Genet. **11**, 547327 (2020)
27. Liu, W., Shen, X., Li, H., Bi, X., Liu, B., Pun, C.M., Cun, X.: Depth-aware test-time training for zero-shot video object segmentation. In: CVPR, pp. 19218–19227 (2024)
28. Liu, W., Shen, X., Pun, C.M., Cun, X.: Explicit visual prompting for low-level structure segmentations. In: CVPR, pp. 19434–19445 (2023)

29. Liu, Z., et al.: Swin transformer: hierarchical vision transformer using shifted windows. In: ICCV, pp. 10012–10022 (2021)
30. Long, J., Shelhamer, E., Darrell, T.: Fully convolutional networks for semantic segmentation. In: CVPR, pp. 3431–3440 (2015)
31. Luo, S., Chen, X., Chen, W., Li, Z., Wang, S., Pun, C.M.: Devignet: high-resolution vignetting removal via a dual aggregated fusion transformer with adaptive channel expansion. In: AAAI (2024)
32. Milletari, F., Navab, N., Ahmadi, S.A.: V-Net: fully convolutional neural networks for volumetric medical image segmentation. In: 3DV, pp. 565–571 (2016)
33. Peng, Y., Yu, D., Guo, Y.: MShNet: multi-scale feature combined with h-network for medical image segmentation. Biomed. Signal Process. Control **79**, 104167 (2023)
34. Ronneberger, O., Fischer, P., Brox, T.: U-Net: convolutional networks for biomedical image segmentation. In: MICCAI, pp. 234–241 (2015)
35. Schlemper, J., et al.: Attention gated networks: learning to leverage salient regions in medical images. Med. Image Anal. **53**, 197–207 (2019)
36. Schuurmans, M., Berman, M., Blaschko, M.B.: Efficient semantic image segmentation with superpixel pooling. arXiv (2018)
37. Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., Liu, Y.: RoFormer: enhanced transformer with rotary position embedding. Neurocomputing **568**, 127063 (2024)
38. Tang, H., et al.: RM-UNet: UNet-like mamba with rotational SSM module for medical image segmentation. Signal, Image and Video Processing (2024)
39. Touvron, H., et al.: LLaMA: open and efficient foundation language models. arXiv (2023)
40. Vaswani, A., et al.: Attention is all you need. In: NeurIPS, vol. 30 (2017)
41. Yang, F., Sun, Q., Jin, H., Zhou, Z.: Superpixel segmentation with fully convolutional networks. In: CVPR, pp. 13964–13973 (2020)
42. Zhang, C., et al.: Correction of out-of-focus microscopic images by deep learning. Comput. Struct. Biotechnol. J. **20**, 1957–1966 (2022)
43. Zhao, C., Zhu, W., Feng, S.: Superpixel guided deformable convolution network for hyperspectral image classification. IEEE TIP **31**, 3838–3851 (2022)
44. Zheng, F., et al.: SMAFormer: Synergistic multi-attention transformer for medical image segmentation. In: BIBM (2024)
45. Zhou, T., Chen, X., Shen, Y., Nieuwoudt, M., Pun, C.M., Wang, S.: Generative AI enables EEG data augmentation for Alzheimer's disease detection via diffusion model. In: ISPCE-ASIA, pp. 1–6 (2023)
46. Zhou, Z., et al.: DocDeshadower: frequency-aware transformer for document shadow removal. arXiv (2023)
47. Zhu, A.Z., et al.: Superpixel transformers for efficient semantic segmentation. In: IROS (2023)